The Five Levels of AI Coding

From Spicy Autocomplete to the Software Factory

Based on Dan Shapiro's framework

Level 0 Manual Labor

> *"Your parents' Volvo"*

- AI used only as an enhanced search tool
- Work progresses at human typing speed
- Full human authorship throughout

L0: Manual Labor

Level 1 Assisted Tasks

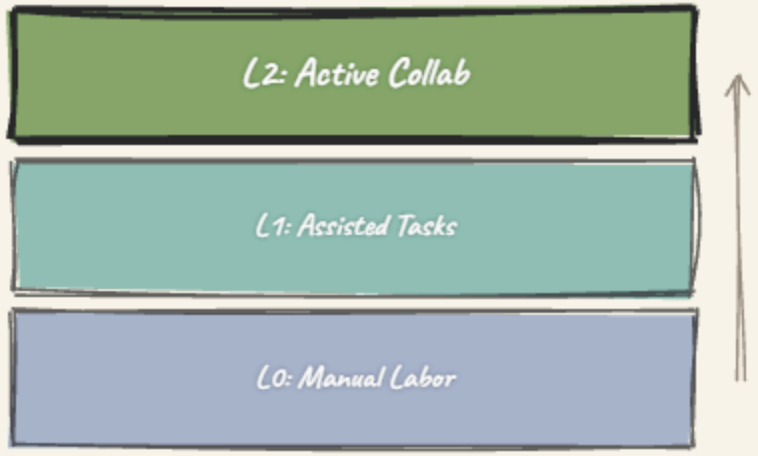
> *"Lane-keeping & cruise control"*

- Discrete delegation: unit tests, docstrings
- Tools: GitHub Copilot, ChatGPT
- Productivity gains without changing job fundamentals

Level 2 Active Collaboration

> *"Highway Autopilot"*

- AI becomes a true colleague — offload tedious work
- Flow state becomes achievable
- **Danger:** feels complete, but it's only intermediate



Level 3 Human Oversight

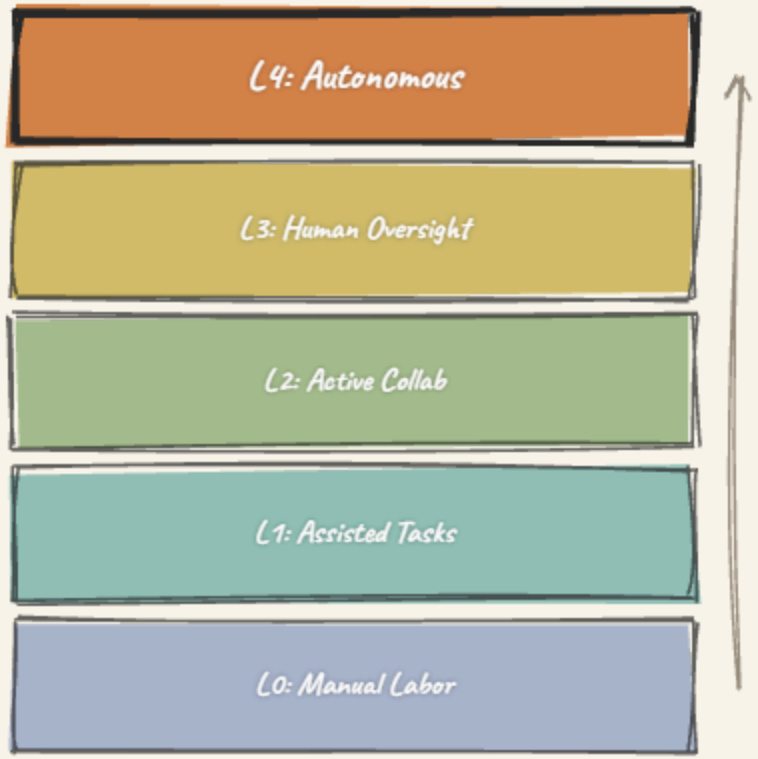
> *"Waymo with a safety driver"*

- Human transitions to managerial role
- Days spent reviewing large code diffs
- Most practitioners plateau here

Level 4 Autonomous Operation

> "Robotaxi"

- Developers write specs — not code
- AI runs autonomously for extended periods
- Humans return later to check test results



Level 5 Fully Automated

> *"The Dark Factory"*

- Specs transform directly into software
- Opaque, automated processes end-to-end
- Teams of <5 running massive systems

Where are you?

Each level demands different tools, mindsets, and workflows.

danshapiro.com · January 2026